

OSALP 2025-04 Phase 2 Summative Assessment

OSALP 2025-04 第二階段總結評估

Financial Risk Management Context

金融風險管理情境

Calculator Allowed; Computer Not Allowed

Total: 25 Problems

可使用計算機；不可使用電腦

共 25 題

General Instructions

一般指示

A calculator may be used for arithmetic, percentages, powers, square roots, and memory operations.

學生可使用計算機處理四則運算、百分比、乘方、平方根及記憶功能。

Computers, phones, spreadsheets, online calculators, programmable tools, and AI tools are not allowed.

不得使用電腦、電話、試算表、網上計算機、可編程工具或人工智能工具。

Students are **not expected to memorise distribution formulas**. If a distribution calculation is required, the question provides a short verbal hint describing the calculation structure.

學生毋須背誦分佈公式。如題目需要進行分佈計算，題目會提供簡短文字提示，說明計算結構。

Unless otherwise stated, ignore taxes, transaction costs, and fees.

除非另有說明，請忽略稅項、交易成本及費用。

Section A. Expected Return, Expected Loss, and Scenario Risk

甲部：預期回報、預期損失及情境風險

Question 1. Back-Substituting a Missing Scenario Return

第 1 題：反推缺失的情境回報率

A student analyst studies a stock currently priced at HK\$50. After applying a three-scenario model, the analyst reports that the **expected stock price after one month** is HK\$51.80.

一名學生分析員研究一隻現價為 HK\$50 的股票。套用三情境模型後，分析員報告該股票一個月後的預期價格為 HK\$51.80。

The three scenarios are as follows: in a strong market, the probability is 30% and the stock return is 12%; in a normal market, the probability is 50% and the stock return is unknown; in a weak market, the probability is 20% and the stock return is -10%.

三個情境如下：在強市況中，概率為 30%，股票回報為 12%；在正常市況中，概率為 50%，股票回報未知；在弱市況中，概率為 20%，股票回報為 -10%。

What is the missing return in the normal market scenario?

在正常市場情境下，缺失的回報率是多少？

A. 2.4%

B. 3.6%

C. 6.0%

D. 4.0%

A. 2.4%

B. 3.6%

C. 6.0%

D. 4.0%

Correct Answer: D

正確答案：D

Calculation Trail:

計算過程：

Expected return:

預期回報率：

$$(51.80 - 50) / 50 = 3.6\%$$

Set up the expected return equation:

建立預期回報率方程：

$$30\%(12\%) + 50\%(x) + 20\%(-10\%) = 3.6\% \quad 30\% (12\%) + 50\% (x) + 20\% (-10\%) = 3.6\%$$

$$3.6\% + 0.5x - 2.0\% = 3.6\% \quad 3.6\% + 0.5x - 2.0\% = 3.6\% \quad 0.5x = 2.0\% \quad 0.5x = 2.0\% \quad x = 4.0\% \quad x = 4.0\%$$

Question 2. Expected Profit with Missing Loss Amount

第 2 題：已知預期利潤下反推缺失的虧損金額

A trading strategy has the following one-month outcomes. All gains and losses below are already net of transaction costs. The analyst reports that the expected profit of the strategy is HK\$6,000.

某交易策略有以下一個月結果。以下所有收益及虧損均已扣除交易成本。分析員報告該策略的預期利潤為 HK\$6,000。

The one-month outcomes are as follows: a large gain occurs with probability 20% and gives +HK\$50,000; a small gain occurs with probability 50% and gives +HK\$10,000; a loss occurs with probability 30% and has an unknown loss amount.

一個月結果如下：大幅獲利的概率為 20%，利潤為 +HK\$50,000；小幅獲利的概率為 50%，利潤為 +HK\$10,000；虧損的概率為 30%，虧損金額未知。

What loss amount is consistent with the reported expected profit?

哪一個虧損金額與所報告的預期利潤一致？

- A. HK\$30,000
- B. HK\$20,000
- C. HK\$40,000
- D. HK\$50,000

- A. HK\$30,000
- B. HK\$20,000
- C. HK\$40,000
- D. HK\$50,000

Correct Answer: A

正確答案：A

Calculation Trail:

計算過程：

$$0.2(50,000) + 0.5(10,000) - 0.3x = 6,000. 0.2(50,000) + 0.5(10,000) - 0.3x = 6,000.$$

$$10,000 + 5,000 - 0.3x = 6,000. 10,000 + 5,000 - 0.3x = 6,000. 15,000 - 6,000 = 0.3x. 15,000 - 6,000 = 0.3x.$$

$$9,000 = 0.3x. 9,000 = 0.3x. x = 30,000. x = 30,000.$$

Question 3. Break-Even Loss from a Given Success Probability

第 3 題：由給定成功概率求打和虧損

A trader estimates a 40% chance of success. If successful, the position gains HK\$20,000. If unsuccessful, it loses an unknown amount.

一名交易員估計成功機會為 40%。若成功，持倉賺取 HK\$20,000；若不成功，則會虧損一個未知金額。

The trader wants the expected profit to be exactly zero. Ignore research time already spent, since it is not part of the payoff calculation.

交易員希望預期利潤剛好為零。請忽略已花費的研究時間，因為它不屬於盈虧計算的一部分。

What is the maximum loss per unsuccessful trade that keeps the expected profit at zero?

每次不成功交易的最高虧損是多少，才能使預期利潤維持在零？

- A. HK\$10,000
- B. HK\$13,333
- C. HK\$20,000
- D. HK\$30,000

- A. HK\$10,000
- B. HK\$13,333
- C. HK\$20,000
- D. HK\$30,000

Correct Answer: B

正確答案：B

Hint: Break-even means probability-weighted gain balances probability-weighted loss.

提示：打和表示經概率加權的收益與經概率加權的虧損互相抵消。

Calculation Trail:

計算過程：

$$40\%(20,000)=60\%(x).40\%(20,000)=60\%(x).8,000=0.6x.8,000=0.6x.$$

$$x=13,333.33.x=13,333.33.$$

So the closest answer is **HK\$13,333**.

因此最接近的答案是 **HK\$13,333**。

Question 4. Risk-Adjusted Judgement Using Downside Ratio

第 4 題：使用下行比率作風險調整判斷

A risk committee compares two strategies. Both have positive expected profit, but their possible losses differ greatly.

一個風險委員會比較兩個策略。兩者均有正的預期利潤，但可能虧損的大小差異很大。

The two strategies are described as follows: Strategy X has expected profit HK\$5,000 and possible loss HK\$20,000; Strategy Y has expected profit HK\$4,000 and possible loss HK\$5,000.

兩個策略如下：策略 X 的預期利潤為 HK\$5,000，可能虧損為 HK\$20,000；策略 Y 的預期利潤為 HK\$4,000，可能虧損為 HK\$5,000。

The committee compares expected profit per HK\$1 of possible loss.

委員會比較每 HK\$1 可能虧損所帶來的預期利潤。

Which statement is MOST appropriate?

哪一項說法最合適？

A. Strategy X is better because its expected profit is higher.

B. Strategy X and Strategy Y are equally attractive because both expected profits are positive.

C. Strategy Y is better on expected profit per dollar of possible loss.

D. Strategy Y is automatically worse because its expected profit is lower.

- A. 策略 X 較好，因為其預期利潤較高。
- B. 策略 X 和策略 Y 同樣吸引，因為兩者的預期利潤均為正。
- C. 就每一元可能虧損的預期利潤而言，策略 Y 較好。
- D. 策略 Y 自動較差，因為其預期利潤較低。

Correct Answer: C

正確答案：C

Calculation Trail:

計算過程：

For Strategy X:

策略 X：

$$5,000/20,000=0.25. 5,000/20,000=0.25.$$

For Strategy Y:

策略 Y：

$$4,000/5,000=0.80. 4,000/5,000=0.80.$$

Strategy Y gives more expected profit per dollar of possible loss.

策略 Y 在每一元可能虧損下提供較高的預期利潤。

Section B. Uniform, Bernoulli, Binomial, and Poisson-Type Risk Models

乙部：均勻、伯努利、二項及泊松類風險模型

Question 5. Uniform Distribution: Recovering the Upper Bound

第 5 題：均勻分佈——反推上限

A student analyst studies the transaction cost of buying a thinly traded stock. The cost rate is equally likely to fall anywhere between 0.2% and an unknown upper bound. The planned trade size is HK\$500,000. The expected transaction cost is HK\$3,000.

一名學生分析員研究買入一隻交投淡薄股票的交易成本。成本率同樣可能落在 0.2% 與一個未知上限之間的任何位置。計劃交易金額為 HK\$500,000。預期交易成本為 HK\$3,000。

The brokerage fee has already been included in the cost range.

經紀費已包含在成本範圍內。

What is the upper bound of the transaction cost rate?

交易成本率的上限是多少？

- A. 1.0%
- B. 0.6%
- C. 0.8%
- D. 1.2%

- A. 1.0%
- B. 0.6%
- C. 0.8%
- D. 1.2%

Correct Answer: A

正確答案：A

Hint: For a uniform range, use the midpoint.

提示：對於均勻範圍，使用中點。

Calculation Trail:

計算過程：

Expected transaction cost rate:

$$3,000/500,000=0.6\%. 3,000/500,000=0.6\%.$$

Since 0.6% is the midpoint:

$$0.2\%+x=0.6\%. \frac{0.2\%+x}{2}=0.6\%. 0.2\%+x=1.2\%. 0.2\%+x=1.2\%. x=1.0\%. x=1.0\%.$$

Question 6. Bernoulli Risk: Recovering the Required Favorable Probability

第 6 題：伯努利風險——反推所需有利概率

A one-day trading position gains HK\$5,000 if the market movement is favorable and loses HK\$4,000 if the market movement is unfavorable.

一個單日交易持倉在市場走勢有利時賺取 HK\$5,000；若市場走勢不利，則虧損 HK\$4,000。

The trader wants the expected profit to be HK\$1,400.

交易員希望預期利潤為 HK\$1,400。

What favorable probability is needed?

需要多大的有利概率？

- A. 40%
- B. 50%
- C. 70%
- D. 60%

- A. 40%
- B. 50%
- C. 70%
- D. 60%

Correct Answer: D

正確答案：D

Hint: Let the favorable probability be p .

提示：設有利概率為 p 。

Calculation Trail:

計算過程：

$$\begin{aligned} 5,000p - 4,000(1-p) &= 1,400. 5,000p - 4,000(1-p) = 1,400. 5,000p - 4,000 + 4,000p = 1,400. 5,000p - \\ 4,000 + 4,000p &= 1,400. 9,000p = 5,400. 9,000p = 5,400. p = 60\%. p = 60\% \end{aligned}$$

Question 7. Binomial Risk: At Least One Losing Day

第 7 題：二項風險——至少一日出現虧損

Each of the next 4 trading days independently has a 20% chance of producing a loss.

未來 4 個交易日中，每一日獨立地有 20% 機會出現虧損。

What is the probability of experiencing **at least one** losing day?

出現至少一個虧損日的概率是多少？

- A. 20.0%
- B. 59.0%
- C. 40.0%
- D. 80.0%

- A. 20.0%
- B. 59.0%
- C. 40.0%
- D. 80.0%

Correct Answer: B

正確答案：B

Hint: Consider the complement event.

提示：考慮補事件。

Calculation Trail:

計算過程：

Probability of no loss in one day:

一天內沒有虧損的概率：

$$100\% - 20\% = 80\%. 100\% - 20\% = 80\%.$$

Probability of no loss over 4 independent days:

連續 4 個獨立交易日都沒有虧損的概率：

$$0.8^4 = 0.4096.$$

Probability of at least one loss:

至少有一天出現虧損的概率：

$$1 - 0.4096 = 0.5904. 1 - 0.4096 = 0.5904.$$

So the probability is approximately **59.0%**.

因此概率約為 59.0%。

Question 8. Expected Number of Breach Days

第 8 題：違規日數的期望值

A portfolio has a 10% chance of breaching its daily loss limit on any trading day. The risk team wants to choose a monitoring window for which the expected number of breach days is 2.

某投資組合在任何交易日有 10% 機會超出每日虧損限額。風險團隊希望選擇一個監察期間，使預期違規日數為 2 日。

Assume the daily probability is the same each day.

How many trading days should be included in the monitoring window?

假設每日概率每天相同。監察期間應包括多少個交易日？

- A. 10
- B. 15
- C. 20
- D. 30

- A. 10
- B. 15
- C. 20
- D. 30

Correct Answer: C

正確答案：C

Hint: Expected number of events equals number of trials multiplied by event probability.

提示：事件的期望次數等於試驗次數乘以事件概率。

Calculation Trail:

計算過程：

Let the number of trading days be n .

設交易日數為 n 。

$$n \times 10\% = 2. n \times 10\% = 2. 0.1n = 2. 0.1n = 2. n = 20. n = 20.$$

Question 9. Poisson-Type Alert Probability over Two Hours

第 9 題：兩小時內的泊松類警報概率

A trading system generates risk alerts. The risk team becomes overloaded if there are 5 or more alerts in an hour. The probability distribution for one hour is given below.

For one hour, the number of alerts has the following probability distribution: 0 alerts has probability 5%; 1 alert has probability 15%; 2 alerts has probability 22%; 3 alerts has probability 22%; 4 alerts has probability 17%; and 5 or more alerts has probability 19%.

一小時內警報數目的概率分佈如下：0 個警報的概率為 5%；1 個警報的概率為 15%；2 個警報的概率為 22%；3 個警報的概率為 22%；4 個警報的概率為 17%；5 個或以上警報的概率為 19%。

Assume two consecutive hours are independent.

What is the probability of overload in at least one of the two hours?

交易系統會產生風險警報。若一小時內有 5 個或以上警報，風險團隊便會超出負荷。下表列出一小時內的概率分佈。假設連續兩個小時互相獨立。兩個小時中至少有一個小時超出負荷的概率是多少？

- A. 34.4%
- B. 19.0%
- C. 38.0%
- D. 81.0%

- A. 34.4%
- B. 19.0%
- C. 38.0%

D. 81.0%

Correct Answer: A

正確答案：A

Hint: Use the complement event over two independent hours.

提示：在兩個獨立小時中使用補事件。

Calculation Trail:

計算過程：

Probability of overload in one hour:

一小時內超出負荷的概率：

$$19\% \text{ or } 19\%$$

Probability of no overload in one hour:

一小時內沒有超出負荷的概率：

$$100\% - 19\% = 81\% \text{ or } 100\% - 19\% = 81\%$$

Probability of no overload in two independent hours:

兩個獨立小時都沒有超出負荷的概率：

$$0.81^2 = 0.6561 \text{ or } 0.81^2 = 0.6561$$

Probability of overload in at least one hour:

至少一個小時超出負荷的概率：

$$1 - 0.6561 = 0.3439 \text{ or } 1 - 0.6561 = 0.3439$$

So the answer is approximately **34.4%**.

因此答案約為 **34.4%**。

Question 10. Poisson-Type Stress Probability: Exactly One

Stressed Hour

第 10 題：泊松類壓力概率——剛好一個小時受壓

A risk desk receives urgent margin calls. The desk is considered stressed if it receives **more than 4 calls** in an hour.

The cumulative probabilities are as follows: the probability of at most 2 calls is 42%; the probability of at most 4 calls is 82%; and the probability of at most 6 calls is 96%.

累積概率如下：最多 2 個通知的概率為 42%；最多 4 個通知的概率為 82%；最多 6 個通知的概率為 96%。

Assume two consecutive hours are independent.

What is the probability that the desk is stressed in **exactly one** of the two hours?

風險交易台收到緊急追繳保證金通知。若一小時內收到超過 4 個通知，該交易台便被視為受壓。假設連續兩個小時互相獨立。該交易台在兩個小時中剛好一個小時受壓的概率是多少？

- A. 18.0%
- B. 32.8%
- C. 67.2%
- D. 29.5%

- A. 18.0%
- B. 32.8%
- C. 67.2%
- D. 29.5%

Correct Answer: D

正確答案：D

Hint: “Exactly one” means stressed in the first hour but not the second, or not stressed in the first hour but stressed in the second.

提示：「剛好一個」表示第一小時受壓而第二小時不受壓，或第一小時不受壓而第二小時受壓。

Calculation Trail:

計算過程：

Stress means more than 4 calls.

受壓表示一小時內收到超過 4 個追繳保證金通知。

$P(\text{stress}) = 100\% - 82\% = 18\%$. $P(\text{no stress}) = 82\%$.
 $P(\text{exactly one stressed hour}) = 2 \times (0.18) \times (0.82) = 0.2952$.

Exactly one stressed hour:

剛好一個小時受壓：

$$2(0.18)(0.82) = 0.2952$$

So the probability is approximately **29.5%**.

因此概率約為 29.5%。

Section C. Portfolio Risk, FX Risk, and Liquidity Risk

丙部：投資組合風險、外匯風險及流動性風險

Question 11. Weighted Portfolio Return with Missing Asset Return

第 11 題：加權投資組合回報與缺失資產回報

A portfolio invests in three assets. The overall portfolio return is reported as 4.6%. The weights already add up to 100%.

The asset weights and returns are as follows: Stock A has weight 50% and return 8%; Stock B has weight 30% and unknown return; Cash has weight 20% and return 0%.

各資產的權重及回報如下：股票 A 的權重為 50%，回報為 8%；股票 B 的權重為 30%，回報未知；現金的權重為 20%，回報為 0%。

What is Stock B's return?

某投資組合投資於三項資產。整體投資組合回報報告為 4.6%。各項權重已合計為 100%。股票 B 的回報是多少？

- A. 1%
- B. 3%
- C. 2%
- D. 4%

- A. 1%
- B. 3%
- C. 2%
- D. 4%

Correct Answer: C

正確答案：C

Calculation Trail:

計算過程：

$$50\%(8\%)+30\%(x)+20\%(0\%)=4.6\%.50\%(8\%)+30\%(x)+20\%(0\%)=4.6\%$$

$$4.0\%+0.3x=4.6\%.4.0\%+0.3x=4.6\%.0.3x=0.6\%.0.3x=0.6\%.x=2\%.x=2\%$$

Question 12. Diversification with Total-Dollar Loss

第 12 題：以整個投資組合金額計算分散投資虧損

A HK\$1,000,000 portfolio invests equally in two assets. In Scenario 2, Asset A returns -8% and Asset B returns 6%.

What is the dollar change of the whole portfolio in Scenario 2?

一個 HK\$1,000,000 的投資組合平均投資於兩項資產。在情境 2 中，資產 A 的回報為 -8%，資產 B 的回報為 6%。在情境 2 中，整個投資組合的金額變動是多少？

- A. Gain of HK\$10,000
- B. Loss of HK\$10,000
- C. Loss of HK\$20,000
- D. Gain of HK\$60,000

- A. 賺取 HK\$10,000
- B. 虧損 HK\$10,000
- C. 虧損 HK\$20,000
- D. 賺取 HK\$60,000

Correct Answer: B

正確答案：B

Calculation Trail:

計算過程：

Equal-weighted portfolio return:

等權重投資組合回報：

$$-8\% + 6\% \div 2 = -1\% \quad \left| \frac{-8\% + 6\%}{2} \right| = -1\%$$

Dollar change:

金額變動：

$$-1\% \times 1,000,000 = -10,000 \quad -1\% \times 1,000,000 = -10,000$$

So the whole portfolio loses **HK\$10,000**.

因此整個投資組合虧損 **HK\$10,000**。

Question 13. Correlation-Like Scenario Interpretation with Hedge Choice

第 13 題：類相關情境解讀與對沖選擇

A risk manager holds Asset X and wants to add another asset that may reduce losses when Asset X performs poorly.

The following scenario returns are estimated:

The estimated scenario returns are as follows: in a good market, Asset X returns 9%, Asset Y returns 8%, and Asset Z returns -2%; in a bad market, Asset X returns -7%, Asset Y returns -6%, and Asset Z returns 3%.

估計情境回報如下：在好市況中，資產 X 回報為 9%，資產 Y 回報為 8%，資產 Z 回報為 -2%；在壞市況中，資產 X 回報為 -7%，資產 Y 回報為 -6%，資產 Z 回報為 3%。

Which conclusion is MOST appropriate?

一名風險經理持有資產 X，並希望加入另一項資產，以便在資產 X 表現欠佳時減少虧損。以下為估計的情境回報。哪一項結論最合適？

A. Asset Y is the better hedge because it earns a high return in the good market.

B. Asset Y removes most of Asset X's risk because its bad-market loss is slightly smaller.

- C. Asset Z is worse because it performs poorly in the good market.
- D. Asset Z may provide better hedging benefit because it rises when Asset X falls.

- A. 資產 Y 是較佳對沖，因為它在好市況中錄得高回報。
- B. 資產 Y 可消除資產 X 的大部分風險，因為它在壞市況中的虧損稍為較小。
- C. 資產 Z 較差，因為它在好市況中表現不佳。
- D. 資產 Z 可能提供較佳對沖效益，因為它在資產 X 下跌時上升。

Correct Answer: D

正確答案：D

Calculation Trail:

計算過程：

In the bad market, Asset X has a negative return of -7%.

在壞市況中，資產 X 的回報為 -7%，屬於負回報。

Asset Y also falls in the bad market, so it may provide limited hedging benefit.

資產 Y 在壞市況中同樣下跌，因此可能只能提供有限的對沖效益。

Asset Z rises by 3% in the bad market, so it may help offset losses from Asset X.

資產 Z 在壞市況中上升 3%，因此可能有助抵銷資產 X 的虧損。

Therefore, Asset Z may provide better hedging benefit.

因此，資產 Z 可能提供較佳的對沖效益。

Question 14. Net Foreign Exchange Exposure with Missing Liability

第 14 題：含缺失負債的淨外匯敞口

A Hong Kong investor owns USD 100,000 of US stocks. The USD weakens from HK\$7.80 to HK\$7.70. The investor reports that the net HKD loss due only to FX movement is HK\$5,000 after considering USD liabilities.

The USD stock value itself does not change.

What amount of USD liabilities does the investor have?

一名香港投資者持有 USD 100,000 的美股。美元由 HK\$7.80 轉弱至 HK\$7.70。考慮美元負債後，投資者報告僅因外匯變動造成的港元淨虧損為 HK\$5,000。美股本身的美元價值不變。投資者有多少美元負債？

- A. USD 0
- B. USD 50,000
- C. USD 95,000
- D. USD 100,000

- A. USD 0
- B. USD 50,000
- C. USD 95,000
- D. USD 100,000

Correct Answer: B

正確答案：B

Calculation Trail:

計算過程：

Exchange rate change:

匯率變動：

$$7.80 - 7.70 = 0.10. 7.80 - 7.70 = 0.10.$$

Net USD exposure:

淨美元敞口：

$$5,000/0.10=50,000.5,000/0.10=50,000.$$

Net exposure equals USD assets minus USD liabilities:

淨敞口等於美元資產減去美元負債：

$$50,000=100,000-\text{liabilities}.50,000=100,000-\text{liabilities}.$$

$$\text{liabilities}=50,000.\text{liabilities}=50,000.$$

Question 15. Combined Stock and FX Return with Missing FX Change

第 15 題：股票與外匯合併回報及缺失外匯變動

A Hong Kong investor buys US stocks. The US stock price rises by 4% in USD terms. However, the USD weakens against the HKD.

The final HKD value of the investment is 98.8% of the original HKD value.

Using multiplication, what was the change in the USD/HKD exchange rate?

一名香港投資者買入美股。美股價格以美元計上升 4%，但美元兌港元轉弱。投資的最終港元價值為原本港元價值的 98.8%。使用乘法計算，USD/HKD 匯率變動是多少？

- A. -5%
- B. -4%
- C. -3%
- D. +1%

- A. -5%
- B. -4%
- C. -3%
- D. +1%

Correct Answer: A

正確答案：A

Calculation Trail:

計算過程：

Final HKD value factor:

最終港元價值倍數：

$$0.988/0.988.$$

US stock value factor:

美股價值倍數：

$$1.04/1.04.$$

Let FX factor be x :

設外匯倍數為 x ：

$$1.04x=0.988/1.04 \quad x=0.988/1.04=0.95 \quad x=0.988/1.04=0.95.$$

So the USD/HKD exchange rate changed by:

因此，USD/HKD 匯率變動為：

$$0.95-1=-5\% \quad 0.95-1=-5\%.$$

Question 16. Liquidity Haircut with Sale-Proceeds Trap

第 16 題：流動性折讓與出售所得陷阱

A fund holds a thinly traded asset. In a stressed sale, it expects to sell the asset at a 15% discount.

The expected sale proceeds are HK\$170,000.

What was the asset's value before the liquidity haircut?

某基金持有一項交投淡薄的資產。在壓力出售情況下，基金預期以 15% 折讓出售該資產。預期出售所得為 HK\$170,000。流動性折讓前，該資產的價值是多少？

- A. HK\$25,500
- B. HK\$170,000
- C. HK\$200,000
- D. HK\$230,000

- A. HK\$25,500
- B. HK\$170,000
- C. HK\$200,000
- D. HK\$230,000

Correct Answer: C

正確答案：C

Calculation Trail:

計算過程：

A 15% discount means the fund receives 85% of the original value.

15% 折讓表示基金收到原價值的 85%。

$$85\% \times x = 170,000 \quad 85\% \times x = 170,000 \quad x = 170,000 / 0.85 = 200,000 \quad x = 170,000 / 0.85 = 200,000.$$

Section D. Law of Large Numbers, Central Limit Theorem, VaR, and Tail Risk

丁部：大數法則、中心極限定理、風險值及尾部風險

Question 17. Law of Large Numbers with Total and Average

Mixed

第 17 題：大數法則——混合總額與平均數

A simple risk model has expected one-day loss of HK\$100 per position. Two students simulate independent positions.

The simulation results are as follows: Student A simulates 10 positions and reports total loss HK\$1,800; Student B simulates 1,000 positions and reports average loss HK\$102.

模擬結果如下：學生 A 模擬 10 個持倉，並報告總虧損為 HK\$1,800；學生 B 模擬 1,000 個持倉，並報告平均虧損為 HK\$102。

Which sample average is closer to the expected loss?

一個簡單風險模型的每個持倉一日預期虧損為 HK\$100。兩名學生模擬獨立持倉。哪一個樣本平均數較接近預期虧損？

- A. Student A, because HK\$1,800 is smaller than Student B's total simulated loss.
- B. Student A, because 10 positions are easier to understand than 1,000 positions.
- C. Both are equally close because both simulations are random.
- D. Student B, because the average loss is HK\$102, only HK\$2 away from HK\$100.

- A. 學生 A，因為 HK\$1,800 小於學生 B 的總模擬虧損。
- B. 學生 A，因為 10 個持倉比 1,000 個持倉較容易理解。
- C. 兩者同樣接近，因為兩次模擬都是隨機的。
- D. 學生 B，因為平均虧損為 HK\$102，與 HK\$100 只相差 HK\$2。

Correct Answer: D

正確答案：D

Hint: Compare average loss per position, not total loss.

提示：比較每個持倉的平均虧損，而不是總虧損。

Calculation Trail:

計算過程：

Student A average:

學生 A 的平均數：

$$1,800/10=180.1,800/10=180.$$

Distance from HK\$100:

與 HK\$100 的距離：

$$180-100=80.180-100=80.$$

Student B average is HK\$102.

學生 B 的平均數為 HK\$102。

Distance from HK\$100:

與 HK\$100 的距離：

$$102-100=2.102-100=2.$$

Student B is closer.

學生 B 較接近。

Question 18. Central Limit Theorem

第 18 題：中心極限定理

A portfolio's individual daily losses are skewed: most days have small gains or losses, but occasionally there are large losses.

A student repeatedly takes samples of 100 trading days and calculates the average daily loss for

each sample.

Which statement is MOST consistent with the Central Limit Theorem?

某投資組合的個別每日虧損呈偏態分佈：大多數日子只有小額收益或虧損，但偶爾會出現大額虧損。一名學生重複抽取 100 個交易日的樣本，並計算每個樣本的平均每日虧損。哪一項說法最符合中心極限定理？

- A. The individual daily losses become normally distributed.
 - B. The sample averages may follow a more regular bell-shaped pattern, even if individual daily losses are skewed.
 - C. Extreme daily losses can no longer occur.
 - D. The average of every 100-day sample must be exactly equal to the true average.
-
- A. 個別每日虧損會變成常態分佈。
 - B. 即使個別每日虧損是偏態分佈，樣本平均數也可能呈現較規則的鐘形模式。
 - C. 極端每日虧損不會再出現。
 - D. 每一個 100 日樣本的平均數必須剛好等於真實平均數。

Correct Answer: B

正確答案：B

Hint: The Central Limit Theorem is about sample averages.

提示：中心極限定理討論的是樣本平均數。

Calculation Trail:

計算過程：

The original individual daily losses may remain skewed.

原本的個別每日虧損仍可能保持偏態分佈。

The Central Limit Theorem concerns the distribution of sample averages. With a sufficiently large sample size, those sample averages may form a more regular bell-shaped pattern, even when the individual observations are not normally distributed.

中心極限定理關注的是樣本平均數的分佈。當樣本數足夠大時，即使個別觀察值不是常態分佈，這些樣本平均數也可能形成較規則的鐘形模式。

Question 19. Normal VaR with Positive Mean

第 19 題：正平均回報下的常態 VaR

A portfolio is worth HK\$800,000. Daily return has average 0.1% and standard deviation 1.5%.

What is the approximate 95% one-day VaR?

某投資組合價值 HK\$800,000。每日回報平均為 0.1%，標準差為 1.5%。近似的 95% 一日 VaR 是多少？

- A. HK\$12,000
- B. HK\$18,200
- C. HK\$19,000
- D. HK\$20,600

- A. HK\$12,000
- B. HK\$18,200
- C. HK\$19,000
- D. HK\$20,600

Correct Answer: C

正確答案：C

Hint: Use 1.65 standard deviations below the average return.

提示：使用低於平均回報 1.65 個標準差的位置。

Calculation Trail:

計算過程：

Downside return threshold:

下行回報門檻：

$$0.1\% - 1.65(1.5\%) = -2.475\% \quad 1.65(1.5\%) = 2.475\%$$

$$0.1\% - 2.475\% = -2.375\%$$

VaR:

風險值：

$$2.375\% \times 800,000 = 19,000$$

Question 20. Multi-Day VaR with Square-Root-of-Time

Scaling

第 20 題：使用平方根時間法則的多日 VaR

A HK\$1,200,000 portfolio has daily return volatility of 1.5%. Assume average return is zero.

What is the approximate 4-day 95% VaR?

某 HK\$1,200,000 投資組合的每日回報波動率為 1.5%。假設平均回報為零。近似的 4 日 95% VaR 是多少？

- A. HK\$59,400
- B. HK\$29,700
- C. HK\$49,500
- D. HK\$118,800

- A. HK\$59,400
- B. HK\$29,700
- C. HK\$49,500
- D. HK\$118,800

Correct Answer: A

正確答案：A

Hint: Use square-root-of-time scaling.

提示：使用平方根時間縮放法則。

Calculation Trail:

計算過程：

4-day volatility:

4 日波動率：

$$1.5\% \times 4 = 1.5\% \times 2 = 3.0\% \quad 1.5\% \times \sqrt{4} = 1.5\% \times 2 = 3.0\%$$

95% downside movement:

95% 下行變動：

$$1.65 \times 3.0\% = 4.95\% \quad 1.65 \times 3.0\% = 4.95\%$$

VaR:

風險值：

$$4.95\% \times 1,200,000 = 59,400 \quad 4.95\% \times 1,200,000 = 59,400$$

Question 21. Back-Substituting Daily Volatility from Multi-Day VaR

第 21 題：由多日 VaR 反推每日波動率

A HK\$1,000,000 portfolio has zero average return. The 9-day 95% VaR is estimated as HK\$99,000.

What is the portfolio's daily volatility?

某 HK\$1,000,000 投資組合的平均回報為零。9 日 95% VaR 估計為 HK\$99,000。該投資組合的每日波動率是多少？

- A. 1.0%
- B. 2.0%
- C. 3.0%
- D. 6.0%

- A. 1.0%
- B. 2.0%
- C. 3.0%
- D. 6.0%

Correct Answer: B

正確答案：B

Hint: Use the 95% multiplier and square-root-of-time scaling in reverse.

提示：反向使用 95% 乘數及平方根時間縮放法則。

Calculation Trail:

計算過程：

9-day VaR percentage:

9 日 VaR 百分比：

$$99,000/1,000,000=9.9\%. 99,000/1,000,000=9.9\%$$

9-day volatility:

9 日波動率：

$$9.9\%/1.65=6.0\%. 9.9\%/1.65=6.0\%$$

Since:

由於：

$9=3, \sqrt{9}=3, 6.0\% = \text{daily volatility} \times 3.6.0\% = \text{daily volatility} \times 3. \text{ daily volatility} = 2.0\%. \text{ daily volatility} = 2.0\%.$

Question 22. Historical VaR versus Tail Average

第 22 題：歷史 VaR 與尾部平均數

A risk manager records 20 daily portfolio losses in HK\$ thousands, sorted from smallest to largest. Larger numbers mean larger losses.

Sorted Daily Losses, HK\$ thousands:

0, 1, 1, 2, 3, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 18, 22, 28, 35, 60

The manager uses a rough 95% historical VaR estimate by taking the **second-largest loss** as the threshold. The manager is **not** calculating the average of the largest losses.

What is the VaR?

一名風險經理記錄 20 個每日投資組合虧損，單位為千港元，並由小至大排列。數值越大表示虧損越大。經理以第二大虧損作為粗略的 95% 歷史 VaR 門檻，而不是計算最大虧損的平均數。VaR 是多少？

- A. HK\$28,000
- B. HK\$60,000
- C. HK\$47,500
- D. HK\$35,000

- A. HK\$28,000
- B. HK\$60,000
- C. HK\$47,500
- D. HK\$35,000

Correct Answer: D

正確答案：D

Hint: Pay attention to threshold versus tail average.

提示：留意門檻值與尾部平均數的分別。

Calculation Trail:

計算過程：

There are 20 observations.

共有 20 個觀察值。

The largest loss is HK\$60,000.

最大虧損為 HK\$60,000。

The second-largest loss is HK\$35,000.

第二大虧損為 HK\$35,000。

Since the question asks for the second-largest loss as the VaR threshold, the VaR is:

由於題目要求以第二大虧損作為 VaR 門檻，因此 VaR 為：

HK\$35,000. |text{HK}\\$35,000}.

Section E. Derivatives, Hedging, and Integrated Risk Management

戊部：衍生工具、對沖及綜合風險管理

Question 23. Forward Hedge Saving with Missing Market Rate

第 23 題：遠期對沖節省金額與缺失市場匯率

A Hong Kong investor must pay USD 100,000 in three months. The investor locks in HK\$7.80 per USD using a forward contract.

At maturity, the hedge saves HK\$12,000 compared with paying at the market exchange rate.

What was the market exchange rate at maturity?

一名香港投資者需要在三個月後支付 USD 100,000。投資者使用遠期合約鎖定每 USD 1 兌 HK\$7.80。到期時，與按市場匯率付款相比，該對沖節省 HK\$12,000。到期時的市場匯率是多少？

- A. HK\$7.68 per USD
- B. HK\$7.80 per USD
- C. HK\$7.92 per USD
- D. HK\$8.00 per USD

- A. 每 USD 1 兌 HK\$7.68
- B. 每 USD 1 兌 HK\$7.80
- C. 每 USD 1 兌 HK\$7.92
- D. 每 USD 1 兌 HK\$8.00

Correct Answer: C

正確答案：C

Calculation Trail:

計算過程：

Forward payment:

遠期合約付款：

$$100,000 \times 7.80 = 780,000. 100,000 \times 7.80 = 780,000.$$

If the hedge saves HK\$12,000, then the market payment would have been:

若對沖節省 HK\$12,000，則按市場匯率付款本來會是：

$$780,000 + 12,000 = 792,000. 780,000 + 12,000 = 792,000.$$

Market exchange rate:

市場匯率：

$$792,000 / 100,000 = 7.92. 792,000 / 100,000 = 7.92.$$

Question 24. Call Option Net Profit with Missing Stock Price

第 24 題：認購期權淨利潤與缺失股價

A trader buys a call option with strike price HK\$50 and premium HK\$4 per share.

At expiry, the option's net profit is HK\$9 per share.

What is the stock price at expiry?

一名交易員買入一份行使價為 HK\$50、期權金為每股 HK\$4 的認購期權。到期時，該期權每股淨利潤為 HK\$9。到期時股價是多少？

- A. HK\$63
- B. HK\$55
- C. HK\$59
- D. HK\$67

- A. HK\$63
- B. HK\$55
- C. HK\$59
- D. HK\$67

Correct Answer: A

正確答案：A

Hint: Net profit is payoff minus premium.

提示：淨利潤等於收益減去期權金。

Calculation Trail:

計算過程：

Net profit is:

淨利潤為：

$$\text{payoff} - 4 = 9. \text{ } \text{payoff} = 13. \text{ } \text{payoff} - 4 = 9. \text{ } \text{payoff} = 13.$$

For a call option with strike price HK\$50:

對於行使價為 HK\$50 的認購期權：

$$S - 50 = 13. \text{ } S - 50 = 13. \text{ } S = 63. \text{ } S = 63.$$

Question 25. Integrated VaR and Hedge Back-Substitution

第 25 題：綜合 VaR 與對沖反推

A student team estimates the one-day adverse scenario loss of a HK\$1,500,000 portfolio using a 5th percentile return of -3.6%.

A hedge costs HK\$9,000. After including both the hedge gain and the hedge cost, the hedged loss is HK\$28,000.

What hedge gain is needed in the adverse scenario?

一個學生小組使用第 5 百分位回報 -3.6% 估計一個 HK\$1,500,000 投資組合的一日不利情境虧損。對沖成本為 HK\$9,000。計入對沖收益和對沖成本後，對沖後虧損為 HK\$28,000。在不利情境

下需要多少對沖收益？

- A. HK\$17,000
- B. HK\$26,000
- C. HK\$35,000
- D. HK\$54,000

- A. HK\$17,000
- B. HK\$26,000
- C. HK\$35,000
- D. HK\$54,000

Correct Answer: C

正確答案：C

Hint: Separate the unhedged loss, hedge gain, and hedge cost.

提示：分開處理未對沖虧損、對沖收益及對沖成本。

Calculation Trail:

計算過程：

Unhedged adverse loss:

未對沖的不利情境虧損：

$$3.6\% \times 1,500,000 = 54,000. \quad 3.6\% \times 1,500,000 = 54,000.$$

After hedging:

對沖後：

$$54,000 - \text{hedge gain} + 9,000 = 28,000. \quad 54,000 - \text{hedge gain} + 9,000 = 28,000. \quad 63,000 - \text{hedge gain} = 28,000. \quad 63,000 - \text{hedge gain} = 28,000. \quad \text{hedge gain} = 35,000. \quad \text{hedge gain} = 35,000.$$

Final Answer Key

最終答案表

Q	Answer	Q	Answer	Q	Answer
1	D	10	D	19	C
2	A	11	C	20	A
3	B	12	B	21	B
4	C	13	D	22	D
5	A	14	B	23	C
6	D	15	A	24	A
7	B	16	C	25	C
8	C	17	D		
9	A	18	B		

Q：題號；Answer：答案

Answer Distribution Check

答案分佈檢查

Answer	Count
A	6
B	6
C	7
D	6

This is now much healthier than the previous version. There is no obvious answer-letter pattern, and D is no longer absent.

這個版本比之前健康得多。答案字母沒有明顯規律，而且 D 不再缺席。

